

Recent Advances in Deep Learning - Spring 2026

Final Examination (Version B) - Answer Key & Grading Rubric

Total Marks: 100

Part I: True/False Questions (10 questions, 1 mark each, Total: 10 marks)

1. ✓ (Weight sharing and local receptive fields are the two core mechanisms by which CNNs drastically reduce the number of parameters compared to fully connected networks.)
2. ✗ (In an ideal state, the generated samples are completely indistinguishable from real samples, and the discriminator can only guess, so the output probability should approach 0.5, not 1.)
3. ✗ (Batch Normalization uses the mean and variance of the current batch during training, but uses the global moving average accumulated over the entire training set during the testing/inference phase.)
4. ✓ (As a lightweight variant of LSTM, GRU merges the forget gate and input gate into an update gate, and combines the cell state with the hidden state.)
5. ✓ (The forward process of the diffusion model is a Markov chain that gradually adds noise according to a predefined variance schedule β_t , with no learnable neural network parameters.)
6. ✗ (The first step of ViT is to split the image into multiple fixed-size non-overlapping patches, and then flatten them before feeding them into the subsequent network.)
7. ✗ (Gradient Clipping is used to solve the "exploding gradient" problem by restricting the maximum norm of the gradient to prevent parameters from diverging; solving vanishing gradients usually relies on residual connections or changing activation functions.)
8. ✓ (When $T > 1$, the probability differences between classes are reduced, and the distribution becomes smoother, which is often used in knowledge distillation to soften labels.)
9. ✓ (During the inference phase, the product of the low-rank matrices can be pre-computed and added to the original weight matrix, i.e., $W_{new} = W_0 + \Delta W$, without adding extra network layers.)
10. ✓ (The receptive field of a 1×1 convolution is exactly 1×1 . It does not change the width and height of the feature map, but it can be used to increase or decrease the dimensionality by setting a different number of convolutional kernels.)

Part II: Multiple Choice Questions (20 questions, 2 marks each, Total: 40 marks)

1. **C. 30×30** (Formula: $O = \frac{I-K+2P}{S} + 1 = \frac{32-3+0}{1} + 1 = 30$)
2. **C. The generator can only produce a limited variety of samples, lacking diversity.**
3. **C. Pooling operations usually contain a large number of weight parameters that need to be updated through backpropagation.** (Pooling layers like Max Pooling or Average Pooling are based on fixed sampling rules and have no learnable parameters.)
4. **B. KL Divergence, used to constrain the distribution of the latent space to approach a standard normal distribution.**
5. **B. Aggregates global image features, and its final output is used for classification tasks.**
6. **B. U-Net architecture (typically with attention mechanisms).**
7. **B. Introduces a hierarchical self-attention mechanism based on Shifted Windows.** (This greatly reduces computational complexity, making it linear with image size rather than quadratic.)
8. **A. To coordinate with the ReLU activation function and maintain stable variance during forward and backward propagation.**
9. **B. Parameter-Efficient Fine-Tuning (PEFT, e.g., LoRA).**
10. **B. The unrolled RNN shares the same set of weight parameters across time steps, making the continuous multiplication of gradients prone to vanishing or exploding.**
11. **B. Slightly lowering the 1 and slightly raising the 0s in the true One-Hot labels, turning them into soft labels.**
12. **B. Pulling closer the feature distances of different augmented views (positive samples) of the same image, and pushing apart the feature distances of different images (negative samples).**
13. **B. Through IO optimization, reducing the read and write operations between the expensive High Bandwidth Memory (HBM) and on-chip SRAM.**
14. **D. Batch Normalization (BN)** (When the Batch Size is too small, the estimation of the mean and variance becomes highly unstable, leading to collapsed BN performance.)
15. **C. The receptive field should cover at least the approximate size of the object to be detected, to integrate sufficient contextual information.**
16. **B. It is usually added dynamically during the attention matrix calculation phase, focusing more on the distance relationship between Tokens.**
17. **A. Guidance (e.g., Classifier-Free Guidance).**
18. **C. Label Smoothing.** (Label smoothing modifies the target label at the loss function level and does not belong to data augmentation in the image pixel space.)

19. **B. In the training phase, the activation values of the retained neurons are scaled by multiplying by $\frac{1}{1-p}$, so that the weight scale does not need to be changed in the test phase.**
20. **B. MAE (Masked Autoencoders).**

Part III: Short Answer & Essay Questions (Open-end, 5 questions, 10 marks each, Total: 50 marks)

1. CNN Receptive Field Calculation and Inductive Bias (10 marks)

- **(1) Receptive Field Calculation (5 marks):**

According to the recursive formula for receptive field: $RF_l = RF_{l-1} + (k_l - 1) \times \prod_{i=1}^{l-1} s_i$

First layer receptive field $RF_1 = k_1 = 3$.

Second layer receptive field $RF_2 = RF_1 + (k_2 - 1) \times s_1 = 3 + (3 - 1) \times 2 = 3 + 4 = 7$.

Answer: One pixel on the second layer's feature map corresponds to a 7×7 receptive field on the original image.

- **(2) Inductive Biases (5 marks):**

Locality: Pixels close to each other in an image are strongly correlated, while those far apart are weakly correlated. CNN kernels only operate on local regions, perfectly fitting this characteristic.

Translation Equivariance: If an object in the input image is translated, the activation response on the feature map will also shift by the same amount in the corresponding location. CNNs achieve this through "weight sharing."

2. Visual Architecture Comparison and Edge Deployment Considerations (10 marks)

- **(1) Core Differences (6 marks):**

Global Information Extraction: CNNs (ResNet) rely on stacking local convolutional kernels to slowly expand the receptive field as the network deepens, capturing global information only at later stages; ViT establishes global connections across all Patches in the very first layer through Self-Attention.

Data Dependency: CNNs possess strong inductive biases (locality, translation equivariance), allowing them to converge easily and perform well on small-scale datasets. ViTs lack this prior knowledge and require massive datasets (e.g., JFT-300M) for the model to learn these spatial relationships itself.

- **(2) Edge Deployment Choice (4 marks):**

Prefer ResNet or a hybrid architecture (like MobileViT).

Reasoning: Physical robots have limited computing power. The Attention mechanism of pure ViT has a computational complexity of $O(N^2)$, resulting in high memory usage and latency. CNNs usually have highly optimized operators (like Depthwise Conv) for edge hardware (NPU/GPU). Ideally, use a hybrid architecture: use CNNs in shallow layers for fast, efficient local feature extraction and downsampling, and lightweight Transformers in deep layers on lower-resolution feature maps for global context.

3. Generative Model Principles Comparison (10 marks)

- **(1) GAN Principles (3 marks):** The Generator (G) is responsible for generating fake images from random noise to fool the discriminator; the Discriminator (D) is responsible for distinguishing whether the input image is real or generated by G. They are trained jointly through a minimax game until reaching Nash equilibrium.
- **(2) Diffusion Principles (4 marks):** The **forward process** is a Markov chain with fixed parameters that gradually adds Gaussian noise to the original image until it becomes pure noise. The **reverse process** trains a neural network (usually a U-Net) to predict and subtract the noise at each step, thereby restoring the pure noise back into a real image.
- **(3) Advantages and Disadvantages (3 marks):**
Advantages: The training process of Diffusion models is based on maximum likelihood estimation, which is very stable and less prone to Mode Collapse like GANs. The diversity and quality of generated samples are generally superior to GANs.
Disadvantages: Extremely slow inference speed. GANs can generate an image with just one forward pass, whereas Diffusion models require dozens or even hundreds of iterative denoising steps to generate a single image.

4. Parameter-Efficient Fine-Tuning Framework: LoRA (10 marks)

- **(1) Mathematical Expression (4 marks):** LoRA freezes the original weights $W_0 \in \mathbb{R}^{d \times k}$ and uses the product of two low-rank matrices to represent the parameter increment. The formula is: $h = W_0x + \Delta Wx = W_0x + BAx$. Here, $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$.
- **(2) Reason for $r \ll \min(d, k)$ (3 marks):** Most weight updates in pre-trained models have a very low "intrinsic dimension." Setting an extremely small rank r (e.g., $r = 8$ or $r = 16$) can reduce the number of trainable parameters by thousands of times while preserving the model's ability to adapt to downstream tasks, saving a massive amount of VRAM.
- **(3) Reason for No Inference Overhead (3 marks):** Before inference deployment, the trained low-rank matrices can be directly added to the original weight matrix: $W_{new} = W_0 + B \cdot A$. During inference, the model still only performs a single matrix multiplication $h = W_{new}x$. The network structure and computational graph remain entirely unchanged, resulting in zero extra latency.

5. Underlying Logic of Normalization Strategies (10 marks)

- **(1) BN Core Formula and Variables (5 marks):**
Formula: $y = \gamma \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta$
Derived from data statistics: Mean μ and variance σ^2 (dependent on the data of the current Batch).
Learnable parameters: Scale factor γ and Shift factor β (optimized via backpropagation).
- **(2) Why Transformers Prefer LN (5 marks):**
Sequence lengths in NLP tasks vary. When using BN, extensive Padding operations are required to align data within the same Batch, which severely biases the calculated mean and variance. Moreover, BN becomes highly unstable if the Batch Size is small. LN performs normalization across all feature dimensions (Channel/Hidden Size) of a

single Token or sample, without needing to compute statistics across samples. Therefore, LN is completely unaffected by Batch Size, variable sequence lengths, or Padding, making it naturally suited for autoregressive or long-sequence modeling in Transformer architectures.